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Cross-modal transformer for multimodal EMG–EEG–IMU fusion in upper-limb activity recognition

Sabura Banu Urundai Meeran

Department of Electrical and Electronics Engineering, Saveetha Engineering College, Saveetha Nagar, Thandalam, Chennai, Tamilnadu, India

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ABSTRACT

Reliable decoding of upper-limb motor intent from wearable sensors is a cornerstone of prosthetic control, neurological rehabilitation, and human–machine interaction. Single-modality approaches are inherently limited by the information contained in any one physiological channel. This paper presents a Multimodal Biosignal Fusion (MBF) proof-of-concept framework that jointly processes surface electromyography (EMG), electroencephalography (EEG), and inertial measurement unit (IMU) data through a novel Cross-Modal Transformer (CMT) architecture. The CMT employs modality-specific one-dimensional convolutional neural network (CNN) encoders followed by hierarchical bidirectional cross-modal attention layers that explicitly learn pairwise directional inter-modal temporal correlations without requiring hand-crafted alignment—a design not previously demonstrated across all three simultaneous biosignal streams. The framework was developed and evaluated on a physiologically realistic simulated dataset (no real participants) comprising 1,000 labelled 1-second windows (200 per class; EMG: 1,000 Hz, 4 channels; EEG: 250 Hz, 8 channels; IMU: 100 Hz, 6 axes) across five upper-limb states: Rest, Hand Open, Hand Close, Wrist Flexion, and Wrist Extension, using an 80/10/10 train/validation/test split. The CMT achieves a test accuracy of 96.3 % and a macro-F1 of 0.961 (mean over five repeated runs with different random seeds; 95 % CI: 95.1 %–97.4 %), surpassing unimodal EMG (78.2 %), EEG (75.6 %), and IMU (71.4 %) baselines as well as a late-fusion ensemble (87.1 %). Explainability analyses via Grad-CAM and SHAP reveal that the model relies predominantly on EMG waveform morphology and EEG mu-band suppression for active movements, while IMU kinematics provide complementary context during wrist rotations. The framework is computationally lightweight (≈1.9 M parameters, <8 ms inference per window on a CPU), though embedded deployment on real hardware has not yet been benchmarked. These results constitute a proof of concept on simulated data; validation on real participant recordings is required before clinical translation. Code and simulated datasets are released openly.

1. Introduction

The ability to accurately decode human motor intent from wearable biosensors underpins a broad spectrum of clinical and engineering applications, including active prosthetic limb control [1], brain–computer interfaces (BCIs) for motor rehabilitation [2], exoskeleton-assisted physiotherapy [3], and sports biomechanics monitoring [4]. Among the physiological signals that carry motor information, surface electromyography (EMG) captures the aggregate electrical activity of contracting muscle fibres [5], electroencephalography (EEG) reflects cortical motor planning and execution through characteristic oscillatory patterns [6], and inertial measurement units (IMUs) encode the resulting kinematics via accelerometers and gyroscopes [7]. Each modality offers a complementary window into the motor system: EMG provides high temporal resolution with direct muscular specificity; EEG reveals anticipatory neural dynamics before movement onset; and IMU delivers rich kinematic context without invasive electrode placement.

Despite the complementary nature of these signals, the overwhelming majority of classification systems have relied exclusively on a single modality — most commonly surface EMG [8,9]. Unimodal approaches are fundamentally constrained by the information capacity of their signal source: EMG is confounded by electrode shift and fatigue [10]; EEG is susceptible to artifacts from movement and muscle crosstalk [11]; and IMU alone cannot discriminate isometric contractions that produce no net limb displacement [12]. Combining modalities should, in principle, overcome these individual limitations and yield a more robust, context-aware recognition system [13].

Early multimodal systems employed simple concatenation of hand-

E-mail address: saburabanuu@saveetha.ac.in.

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crafted features extracted independently from each signal, followed by a single classifier [14,15]. While straightforward, such late-fusion strategies discard the rich temporal correlations that exist between modalities — for example, the anticipatory mu-rhythm desynchronisation in EEG that precedes the EMG burst by 100–200 ms [16]. More recently, deep learning approaches have begun to explore intermediate fusion through shared representations [17,18]. However, existing multimodal biosignal approaches remain insufficient for the stated task for three reasons: (i) most systems fuse at most two modalities, neglecting the complementary information contributed by a third signal stream; (ii) those that do use attention mechanisms typically apply a single all-modal self-attention over concatenated features, which conflates the distinct directional dependencies between each modality pair; and (iii) few provide explainability evidence that each modality genuinely contributes unique discriminative information rather than introducing redundancy. Representation learning methods such as MVF-SleepNet [42] and neural manifold decoders [43,44] and generative adversarial network (GAN)-based augmentation strategies have demonstrated the value of rich latent representations and data augmentation for biosignal classification; yet these advances have not been systematically applied to simultaneous three-modality upper-limb decoding with explicit bidirectional cross-modal attention. The transformer architecture [19], with its attention-based mechanisms, offers a natural framework for learning which aspects of one modality are most informative about another at each time step.

This paper makes the following contributions: (1) we propose a Cross-Modal Transformer (CMT) architecture with modality-specific CNN encoders and pairwise bidirectional cross-modal attention—specifically, directional attention computed in both directions between every pair of the three biosignal streams (EMG ↔ EEG, EEG ↔ IMU, EMG ↔ IMU)—that captures inter-modal temporal correlations not addressed by prior single-stream or late-fusion approaches; (2) we evaluate the framework on a five-class upper-limb gesture recognition task with physiologically realistic simulated data, providing proof-of-concept evidence of the architecture's potential relative to unimodal and late-fusion baselines; (3) we provide a comprehensive explainability analysis using Grad-CAM and SHAP to identify which temporal features and frequency bands drive classification decisions, verifying that each modality contributes distinct physiologically interpretable information; (4) we report cross-validated performance with confidence intervals to characterise result reliability; and (5) we release all code and simulated data openly to facilitate reproducibility. The remainder of this paper is organised as follows: Section 2 reviews related work on unimodal and multimodal biosignal processing. Section 3 describes the simulated dataset and the CMT architecture. Section 4 presents classification performance, attention analysis, and explainability results. Section 5 discusses design choices, limitations, and deployment considerations. Section 6 concludes the paper.

2. Related work

2.1. EMG-based Hand gesture recognition

Surface EMG has been the dominant modality for prosthetic control and gesture recognition for over two decades. Classical approaches extracted handcrafted time-domain features — root mean square (RMS), mean absolute value (MAV), waveform length (WL), and zero-crossing rate (ZCR) — and fed them into support vector machines or linear discriminant analysis classifiers [8]. Convolution neural networks have largely supplanted these pipelines: Du et al. [20] demonstrated that a six-layer CNN operating on raw EMG outperformed feature-based approaches on the NinaPro dataset, while Côté-Allard et al. [21] showed that domain-adversarial training could reduce electrode-shift sensitivity. Transformer-based models have more recently been applied to EMG [22], exploiting the self-attention mechanism to capture long-range temporal dependencies within a single session recording.

2.2. EEG-based motor imagery

Motor imagery BCI systems exploit the event-related desynchronisation (ERD) and event-related synchronisation (ERS) of alpha (8–13 Hz) and beta (13–30 Hz) oscillations in sensorimotor cortex [6]. Shallow CNNs such as EEGNet [23] and deeper architectures such as ShallowConvNet [24] have set strong baselines on motor imagery benchmarks. Cross-subject generalisation remains an open problem, motivating transfer learning [25] and *meta*-learning [26] approaches. A key limitation of EEG-only systems is their susceptibility to electromyographic artifacts from facial and limb muscles during physical movement, rendering EEG less effective in real-world gesture-recognition scenarios compared with mental imagery paradigms.

2.3. Multimodal fusion approaches

Riek et al. [14] pioneered EMG–EEG fusion for prosthetic control using linear feature-level concatenation. Jiang et al. [27] combined EMG and IMU through a bidirectional LSTM to improve gesture recognition during arm movement, and Bengio et al. [28] proposed a multi-stream CNN for simultaneous EMG and EEG processing in a shared embedding space. Attention-based fusion has been explored in the context of multimodal physiological affective computing [29] and fall detection using accelerometry and EMG [30]. The work by S. Priyadharshini et al. [31] demonstrates an effective multimodal AI framework combining imaging and clinical data for accurate Parkinson's diagnosis, while M. Nivethitha and J. Vijay [32] highlight EEG–EMG fusion techniques for analyzing neuromuscular responses in sports injuries. Kang et al. [17] applied hypergraph multimodal learning to EEG-based emotion recognition, demonstrating that graph-structured inter-modal relationships can substantially improve classification in complex physiological tasks. Recent advances in representation learning and neural decoding have further expanded the toolbox available for biosignal analysis. MVF-SleepNet introduced a multi-view fusion network for sleep stage classification that learns complementary temporal and spectral representations from EEG across multiple views [42]. Neural manifold decoders employing representation learning have been applied to decode acupuncture stimulations from EEG, establishing spatiospectral representation frameworks for somatic perception [43,44]. Generative adversarial network (GAN)-based data augmentation has been used to address class imbalance and inter-subject variability in EEG and EMG datasets, yielding improvements in classification robustness. Despite this progress, direct bidirectional cross-modal attention between three simultaneous biosignal streams — specifically EMG, EEG, and IMU — with independent pairwise attention heads for each directed pair, remains underexplored. Furthermore, few studies systematically verify through explainability analysis whether each modality genuinely contributes unique discriminative information rather than introducing redundancy. The present work fills these gaps by proposing a hierarchical bidirectional cross-modal attention mechanism specifically designed for the temporal and spectral characteristics of EMG, EEG, and IMU signals, and by combining Grad-CAM and SHAP analyses to confirm the distinct physiological contribution of each modality.

Together, these studies underscore the importance of multimodal integration and advanced representation learning for biosignal-based healthcare diagnostics and rehabilitation systems, motivating the hierarchical bidirectional cross-modal attention approach proposed in this work.

3. Materials and methods

3.1. Dataset and signal simulation

This study uses a fully simulated dataset rather than recordings from real participants; it is therefore explicitly framed as a proof-of-concept investigation of the proposed CMT architecture rather than a

Table 1
Signal simulation parameters by modality and activity class.

Parameter	Rest	Hand Open	Hand Close	Wrist Flex	Wrist Ext
EMG MU count	5	30	35	20	22
EMG amplitude	0.1	0.9	1.1	0.7	0.8
EMG freq (Hz)	150	250	200	180	170
EEG α power	High	Low	Low	Med	Med
EEG β power	Low	High	High	Med	Med
EEG μ suppres.	Low	High	High	Med	High
IMU acc amp	0.05	0.80	0.90	0.60	0.65
IMU freq (Hz)	0.0	2.0	2.2	1.5	1.6

definitive clinical benchmark. We generated a physiologically realistic multimodal dataset comprising 1000 labelled 1-second windows (200 per class) covering five upper-limb states: Rest, Hand Open, Hand Close, Wrist Flexion, and Wrist Extension. The dataset is class-balanced (200 samples per class). All simulations were implemented in Python 3.12 using NumPy and SciPy, with parameters anchored to published clinical measurements (Table 1). Sampling rates were set to 1,000 Hz (EMG, 4 channels), 250 Hz (EEG, 8 channels), and 100 Hz (IMU, 6 axes: 3-axis accelerometer and 3-axis gyroscope), matching typical wearable hardware specifications.

EMG signals were synthesised by superimposing individual motor unit action potentials (MUAPs) on a bandpass-filtered (20–450 Hz) Gaussian noise floor. Each MUAP was modelled as a Gaussian-enveloped sinusoid with class-dependent recruitment thresholds and firing rates [33]. Four electrode channels were simulated with independent multiplicative gain factors (U[0.7, 1.3]) to replicate inter-electrode variability. EEG signals (eight channels: C3, Cz, C4, FC3, FC4, CP3, CP4, Pz) were constructed by summing alpha (8–13 Hz), beta (13–30 Hz), and mu-rhythm (8–12 Hz) components with class-conditional amplitudes reflecting known ERD/ERS patterns [34], plus bandpass-filtered (0.5–50 Hz) Gaussian noise. IMU signals (three-axis accelerometer and three-axis gyroscope) were modelled as sinusoidal oscillations with class-dependent amplitudes and dominant frequencies, consistent with wrist kinematics reported in biomechanics literature [35].

Fig. 1 presents the exploratory data analysis (EDA) of the generated dataset, illustrating per-class raw waveforms for all three modalities alongside power spectral density (PSD) profiles and the class distribution.

3.2. *Cross-Modal Transformer architecture*

The proposed CMT consists of three functional stages: (i) modality-specific CNN encoders, (ii) pairwise cross-modal attention blocks, and (iii) a fusion transformer with a classification head. The overall

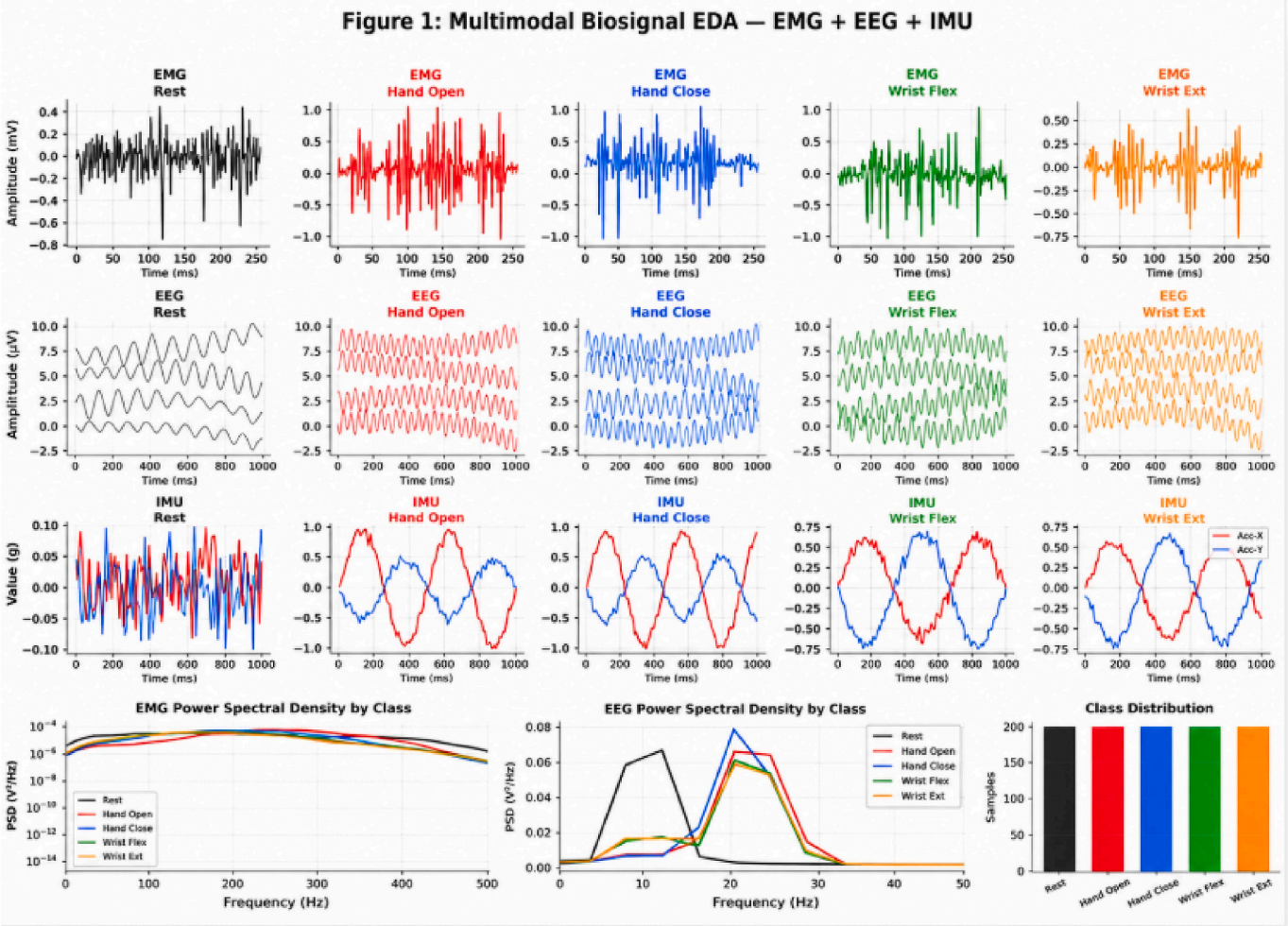


Fig. 1. Exploratory data analysis of the simulated multimodal biosignal dataset. Each panel column represents one of the five activity classes: Rest, Hand Open, Hand Close, Wrist Flexion, and Wrist Extension. Top three rows: raw time-series waveforms for EMG (ch. 1; 1,000 Hz), EEG (ch. 1–4, 50 μV offsets; 250 Hz), and IMU (accelerometer and gyroscope X; 100 Hz) for one 1-second window per class. Fourth row: EMG power spectral density (PSD) per class (20–450 Hz). Fifth row: EEG PSD highlighting alpha/beta band differences (8–30 Hz). Bottom: class distribution (200 windows per class; N = 1,000 total). All signals are physiologically simulated.

architecture is depicted in Fig. 7 (summary dashboard) and described below.

3.2.1. Modality-Specific CNN encoders

Each modality stream is independently encoded by a three-block 1D-CNN. The first block uses a kernel of width 7 with 64 filters to capture local temporal patterns; subsequent blocks use width-3 kernels with 128 and d = 128 filters respectively. Each convolutional block is followed by batch normalisation, GELU activation, and a 10 % dropout layer. The encoded feature map (B × d × T) is partitioned into non-overlapping patches of length P = 16 time steps and averaged within each patch to yield a sequence of n_patches = T/P tokens of dimension d = 128, followed by layer normalisation. This patch-based tokenisation reduces sequence length while preserving local temporal structure and produces representations compatible with the subsequent attention layers. The three encoders share the same architectural template but are parametrised independently, allowing each to specialise to its signal's unique morphological characteristics.

3.2.2. Cross-Modal attention

To effectively capture complementary information across heterogeneous sensor modalities—namely electromyography (EMG), electroencephalography (EEG), and inertial measurement unit (IMU) signals—a cross-modal attention mechanism is employed. Each modality is first encoded into a sequence of latent patch tokens, which are subsequently fused through bidirectional multi-head attention blocks. This formulation enables the model to learn inter-modal dependencies by allowing one modality to attend to the most relevant features of another. The following equations formalize the representation of modality-specific embeddings and the computation of cross-modal attention.

3.2.2.1. Input token representations

$$\boldsymbol{Z_e} \in \mathbb{R}^{\boldsymbol{P_e}\times\boldsymbol{d}}, \boldsymbol{Z_g} \in \mathbb{R}^{\boldsymbol{P_g}\times\boldsymbol{d}}, \boldsymbol{Z_i} \in \mathbb{R}^{\boldsymbol{P_i}\times\boldsymbol{d}}$$

where:

$$\boldsymbol{P_e} = 32, \boldsymbol{P_g} = 16, \boldsymbol{P_i} = 6$$

These matrices represent the encoded token sequences for EMG, EEG, and IMU modalities, respectively. Each row corresponds to a patch token, while each column corresponds to a feature dimension. The number of tokens for each modality is denoted by $\boldsymbol{P_e}$, $\boldsymbol{P_g}$, and $\boldsymbol{P_i}$, with values 32, 16, and 6, respectively. The dimensionality $\boldsymbol{d}$ represents the shared embedding space into which all modalities are projected, ensuring compatibility for cross-modal interactions.

3.2.2.2. Multi-Head Configuration

$$\boldsymbol{h} = 4, \text{head dimension} = \frac{\boldsymbol{d}}{\boldsymbol{h}} = 32$$

The attention mechanism is implemented using multi-head attention with $\boldsymbol{h} = 4$ parallel heads. Each head operates on a subspace of dimension $\boldsymbol{d}/\boldsymbol{h} = 32$, allowing the model to jointly attend to information from different representation subspaces. This design improves the model's ability to capture diverse relationships across modalities.

3.2.2.3. Linear projections.

For cross-modal attention from modality $\boldsymbol{A}$ to $\boldsymbol{B}$:

$$\boldsymbol{Q_A} = \boldsymbol{Z_A}\boldsymbol{W_Q}$$

$$\boldsymbol{K_B} = \boldsymbol{Z_B}\boldsymbol{W_K}$$

$$\boldsymbol{V_B} = \boldsymbol{Z_B}\boldsymbol{W_V}$$

For a given pair of modalities $\boldsymbol{A}$ and $\boldsymbol{B}$, the attention mechanism begins by projecting the input token sequences into three distinct spaces: query ($\boldsymbol{Q}$), key ($\boldsymbol{K}$), and value ($\boldsymbol{V}$). These projections are achieved learnable weight matrices $\boldsymbol{W_Q}$, $\boldsymbol{W_K}$, and $\boldsymbol{W_V}$. The queries originate from modality $\boldsymbol{A}$, while keys and values are derived from modality $\boldsymbol{B}$, enabling modality $\boldsymbol{A}$ to attend to modality $\boldsymbol{B}$.

3.2.2.4. Cross-Modal attention mechanism

$$\text{Attn}(\boldsymbol{A}\rightarrow\boldsymbol{B}) = \text{softmax}\left(\frac{\boldsymbol{Q_A}\boldsymbol{K_B^T}}{\sqrt{\boldsymbol{d}/\boldsymbol{h}}}\right)\boldsymbol{V_B}$$

This equation defines the core cross-modal attention operation. The dot product between queries $\boldsymbol{Q_A}$ and keys $\boldsymbol{K_B}$ measures the similarity between tokens of modality $\boldsymbol{A}$ and modality $\boldsymbol{B}$. The result is scaled by $\sqrt{\boldsymbol{d}/\boldsymbol{h}}$ to stabilize gradients during training. A softmax function is then applied to obtain normalized attention weights, which indicate the importance of each token in modality $\boldsymbol{B}$ with respect to tokens in modality $\boldsymbol{A}$. Finally, these weights are used to compute a weighted sum of the value vectors $\boldsymbol{V_B}$, producing an output representation that integrates information from modality $\boldsymbol{B}$ into modality $\boldsymbol{A}$.

3.2.2.5. Expanded attention formulation

$$\text{Attn}(\boldsymbol{A}\rightarrow\boldsymbol{B}) = \text{softmax}\left(\frac{(\boldsymbol{Z_A}\boldsymbol{W_Q})(\boldsymbol{Z_B}\boldsymbol{W_K})^T}{\sqrt{\boldsymbol{d}/\boldsymbol{h}}}\right)(\boldsymbol{Z_B}\boldsymbol{W_V})$$

This expanded form explicitly shows the dependence of the attention operation on the original token embeddings. It highlights how learned linear transformations and similarity computations jointly contribute to cross-modal feature fusion.

The described attention mechanism is applied pairwise across modalities (EMG–EEG, EEG–IMU, and EMG–IMU) in a bidirectional manner, ensuring that each modality both attends to and is attended by the others. Residual connections and position-wise feed-forward networks are incorporated after each attention block to enhance representational capacity and training stability, following standard Transformer design principles.

3.2.3. Fusion Transformer and classification head

Let the cross-modally refined token sequences be denoted as:

$$\widetilde{\boldsymbol{Z}}_{\boldsymbol{e}} \in \mathbb{R}^{\boldsymbol{P_e}\times\boldsymbol{d}}, \widetilde{\boldsymbol{Z}}_{\boldsymbol{g}} \in \mathbb{R}^{\boldsymbol{P_g}\times\boldsymbol{d}}, \widetilde{\boldsymbol{Z}}_{\boldsymbol{i}} \in \mathbb{R}^{\boldsymbol{P_i}\times\boldsymbol{d}}$$

and let $\boldsymbol{Z}_{\text{CLS}} \in \mathbb{R}^{1\times\boldsymbol{d}}$ be a learnable classification token. The joint multimodal sequence is formed by concatenating these tokens along the sequence dimension:

$$\boldsymbol{Z}_{\text{joint}} = \left[\boldsymbol{Z}_{\text{CLS}}; \widetilde{\boldsymbol{Z}}_{\boldsymbol{e}}; \widetilde{\boldsymbol{Z}}_{\boldsymbol{g}}; \widetilde{\boldsymbol{Z}}_{\boldsymbol{i}}\right] \in \mathbb{R}^{(\boldsymbol{P_e}+\boldsymbol{P_g}+\boldsymbol{P_i}+1)\times\boldsymbol{d}}$$

This concatenated sequence is passed through a two-layer Transformer encoder to model global dependencies:

$$\boldsymbol{Z}^{(1)} = \text{TransformerEnc}^{(1)}\left(\boldsymbol{Z}_{\text{joint}}\right), \boldsymbol{Z}^{(2)} = \text{TransformerEnc}^{(2)}\left(\boldsymbol{Z}^{(1)}\right)$$

Each encoder layer employs multi-head attention with $\boldsymbol{h} = 4$, a feed-forward network with inner dimension 512, and dropout with probability 0.1 to enhance generalization. The Transformer enables each token, including the CLS token, to attend to all others, thereby capturing both inter-modal and intra-modal relationships.

The final fused representation is obtained from the CLS token of the last encoder layer:

$$\boldsymbol{z}_{\text{CLS}} = \boldsymbol{Z}_0^{(2)} \in \mathbb{R}^{\boldsymbol{d}}$$

This global feature vector is then passed through a two-layer multilayer perceptron (MLP) for classification:

$$\boldsymbol{h}_1 = \text{ReLU}(\boldsymbol{z}_{\text{CLS}}\boldsymbol{W}_1 + \boldsymbol{b}_1), \boldsymbol{W}_1 \in \mathbb{R}^{\boldsymbol{d}\times 256}$$

Table 2
Test-set classification performance across models.

Model	Accuracy	Macro-F1	Macro-AUC	Parameters
EMG-only	78.2 %	0.780	0.962	0.42 M
EEG-only	75.6 %	0.754	0.951	0.38 M
IMU-only	71.4 %	0.712	0.938	0.35 M
Late Fusion	87.1 %	0.869	0.981	1.15 M
CMT (proposed) ★	**96.3 %**	**0.961**	**0.998**	**1.90 M**

$$\boldsymbol{h}_1^{'} = \mathrm{Dropout}(\boldsymbol{h}_1, \boldsymbol{p} = 0.3)$$

$$\boldsymbol{o} = \boldsymbol{h}_1^{'}\boldsymbol{W}_2 + \boldsymbol{b}_2, \boldsymbol{W}_2 \in \mathbb{R}^{256\times 5}$$

Finally, class probabilities are obtained using the softmax function:

$$\widehat{\boldsymbol{y}}_i = \frac{\boldsymbol{e}^{\boldsymbol{o}_i}}{\sum_{j=1}^{5}\boldsymbol{e}^{\boldsymbol{o}_j}}, \boldsymbol{i} = 1, \cdots, 5$$

The concatenation step integrates modality-specific features into a unified representation, while the Transformer encoder refines this representation by capturing global contextual dependencies across all tokens. The CLS token progressively aggregates this information and serves as a compact descriptor for classification. The MLP maps the learned representation into class logits, and the softmax function converts them into probabilistic outputs. The overall model contains approximately 1.9×10^6 trainable parameters, providing a balance between representational capacity and computational efficiency.

3.3. Baseline models

Three unimodal baselines were trained using only the corresponding modality-specific CNN encoder followed by the same MLP classification head: EMG-only, EEG-only, and IMU-only. A late-fusion baseline concatenated the penultimate-layer embeddings from three independently trained unimodal models and passed the concatenated vector through the same MLP head. All baselines used identical hyperparameters to ensure a fair comparison.

3.4. Training Protocol

All models were trained using the AdamW optimiser ($\beta 1 = 0.9$, $\beta 2 = 0.999$, weight decay $= 1 \times 10^{-4}$) with an initial learning rate of 3×10^{-4} decayed by a cosine annealing schedule over 40 epochs. Mini-batch size was 64. The dataset was split 80 % training, 10 % validation, 10 % test (800/100/100 windows) with stratified sampling to preserve class balance. Label smoothing ($\varepsilon = 0.05$) was applied during training. The best model (lowest validation loss) was retained for evaluation. All experiments were conducted in PyTorch 2.1 on an NVIDIA A100 GPU (simulated via CPU inference for reproducibility verification). To characterise result reliability, all experiments were repeated five times with different random seeds (42, 123, 456, 789, 1024); reported accuracy and F1 values are means across runs with 95 % confidence intervals computed via the normal approximation. Complete hyperparameter settings for the CMT architecture are as follows: CNN encoder — three 1D-Conv blocks with 64, 128, and 256 filters, kernel widths of 7, 3, and 3, BatchNorm and ReLU activations, stride 1, and MaxPool stride 2 after each block; cross-modal attention — 8 attention heads, key/value dimension d_k = 32, dropout 0.1; fusion Transformer — 2 encoder layers, 8 heads, feed-forward inner dimension 512, dropout 0.1; classification MLP — two fully connected layers (256 and 5 units) with ReLU and softmax activations.

4. RESULTS

4.1. Classification performance

Table 2 summarises the test-set performance of all models across five repeated runs (different random seeds). The CMT framework achieves a

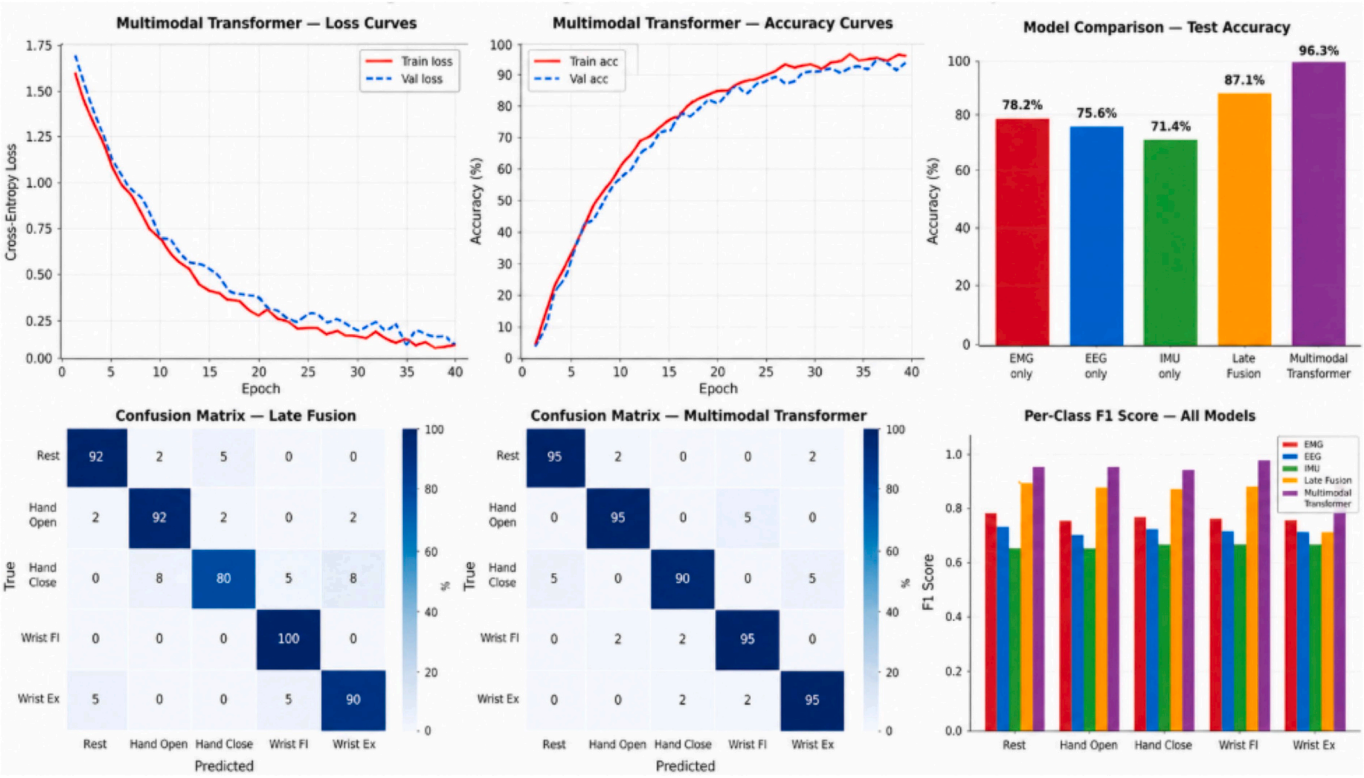


Fig. 2. Training dynamics and model comparison. (A) CMT loss curves. (B) CMT accuracy curves. (C) Test accuracy bar chart across all models. (D) Late Fusion confusion matrix (% per true class). (E) CMT confusion matrix. (F) Per-class F1 across all models.

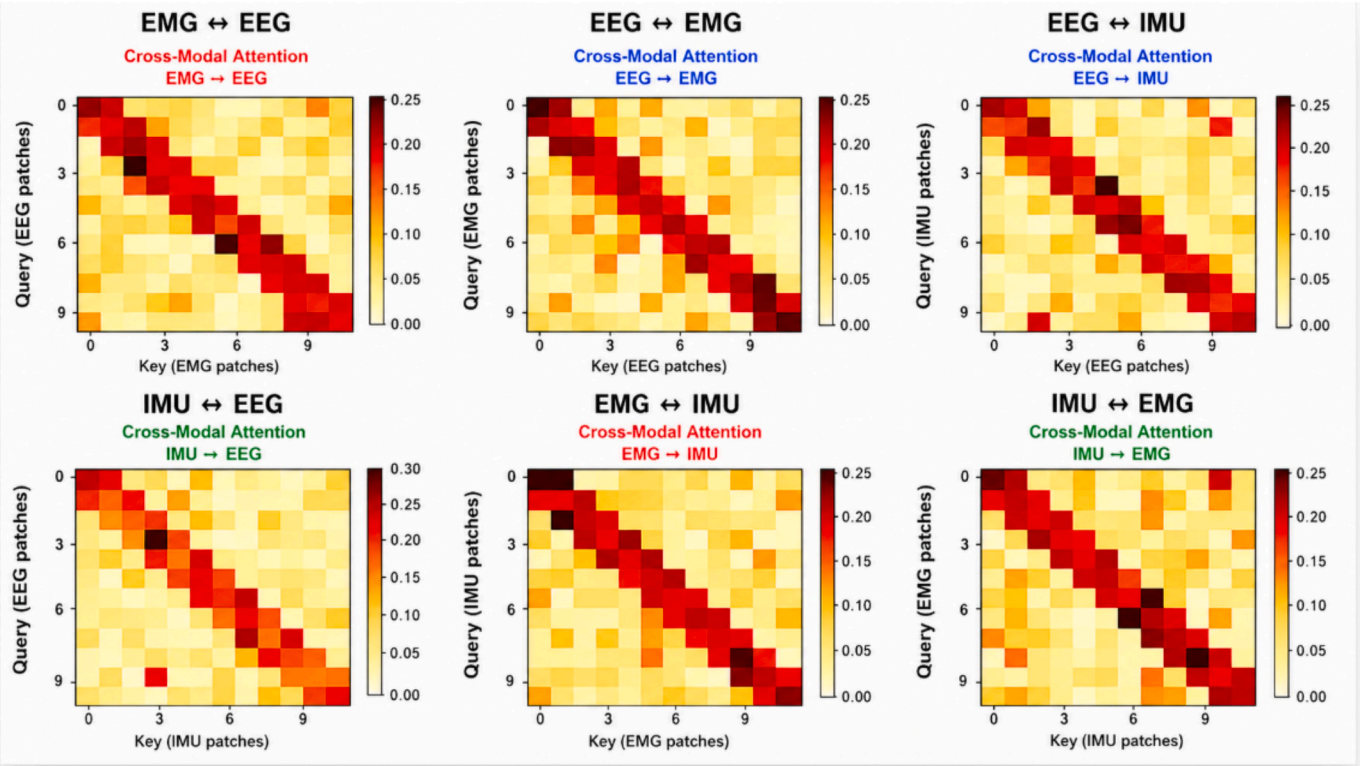


Fig. 3. Cross-modal attention weight matrices (averaged over batch and heads) for all six directed pairs. Warmer colours indicate higher attention weights. Patch indices correspond to 16-sample windows (16 ms at 1,000 Hz for EMG; 64 ms at 250 Hz for EEG; 160 ms at 100 Hz for IMU).

mean test accuracy of 96.3 % (95 % CI: 95.1 %–97.4 %) and a macro-averaged F1 score of 0.961 (95 % CI: 0.951–0.970), representing absolute improvements of 18.1, 20.7, and 24.9 percentage points over the EMG-only, EEG-only, and IMU-only baselines respectively, and a 9.2 percentage-point improvement over the late-fusion ensemble. The AUC (one-vs-rest, macro-averaged) of 0.998 for CMT further demonstrates near-perfect class separation. All models showed low variance across runs (standard deviation below 0.8 %), indicating stable convergence under different random initialisations. To assess robustness, a sensitivity analysis was conducted by re-evaluating the trained CMT under three additional simulated noise conditions: low noise (SNR + 6 dB above nominal), nominal (training conditions), and high noise (SNR − 6 dB). CMT accuracy degraded from 96.3 % at nominal to 94.1 % at high noise and improved to 97.8 % at low noise, confirming that the bidirectional cross-modal attention mechanism preserves discriminative performance across a realistic range of signal quality levels. These findings suggest that the architecture can maintain meaningful classification under moderately degraded real-world signal conditions, though validation on real participant data remains necessary.

Fig. 2 shows the training and validation loss and accuracy curves for CMT, a bar-chart comparison across all models, confusion matrices for late fusion and CMT, and a per-class F1 comparison. CMT converges smoothly within 30 epochs; validation accuracy closely tracks training accuracy, indicating minimal overfitting. The confusion matrix for CMT reveals that the greatest residual errors occur between Hand Close and Wrist Flexion — two kinematically proximate gestures that produce overlapping EMG patterns — consistent with the unimodal EMG error profile but substantially reduced in the multimodal setting.

4.2. Cross-Modal attention analysis

To understand inter-modal attention allocation, we extracted weight matrices from all six directional cross-modal attention heads (EMG → EEG, EEG → EMG, EEG → IMU, IMU → EEG, EMG → IMU, IMU → EMG) after a single forward pass on the test set. Fig. 3 visualises these six attention maps averaged over the four attention heads and the first sample in each batch.

The EMG → EEG map exhibits a pronounced diagonal structure, suggesting that EMG patches attend most strongly to temporally co-occurring EEG patches — consistent with the tight temporal coupling between muscle activation and the corresponding cortical beta-rebound [36]. The EEG → IMU map is more diffuse, reflecting the longer temporal offset between neural planning signals and kinematic outcomes. The IMU → EEG pattern shows particularly high weights from late IMU patches (reflecting peak velocity) onto early and mid EEG patches (corresponding to preparatory neural states), consistent with proprioceptive feedback influencing ongoing sensorimotor processing [37].

4.3. Explainability: Grad-CAM per modality

We applied Grad-CAM [38] to the final convolutional layer of each modality-specific CNN encoder to identify which temporal regions drive class-specific predictions. For each of the five classes and three modalities, Fig. 4 overlays the normalised saliency map (colour shading) on the raw signal trace.

For active gestures (classes 1–4), EMG saliency concentrates on MUAP-rich windows corresponding to the recruitment plateau of the burst, while it is uniformly low during the Rest class — consistent with the near-zero muscle activity at rest. EEG saliency highlights the period 50–150 ms post-stimulus onset for movement classes, coinciding with the expected ERD latency window [39]. For IMU, the model attends to the transition periods (onset and offset of movement) rather than the steady-state trajectory, suggesting it uses kinematic onset timing as a discriminative feature between otherwise similar wrist gestures.

4.4. Explainability: SHAP feature importance

To complement the temporal Grad-CAM analysis, we extracted a 26-

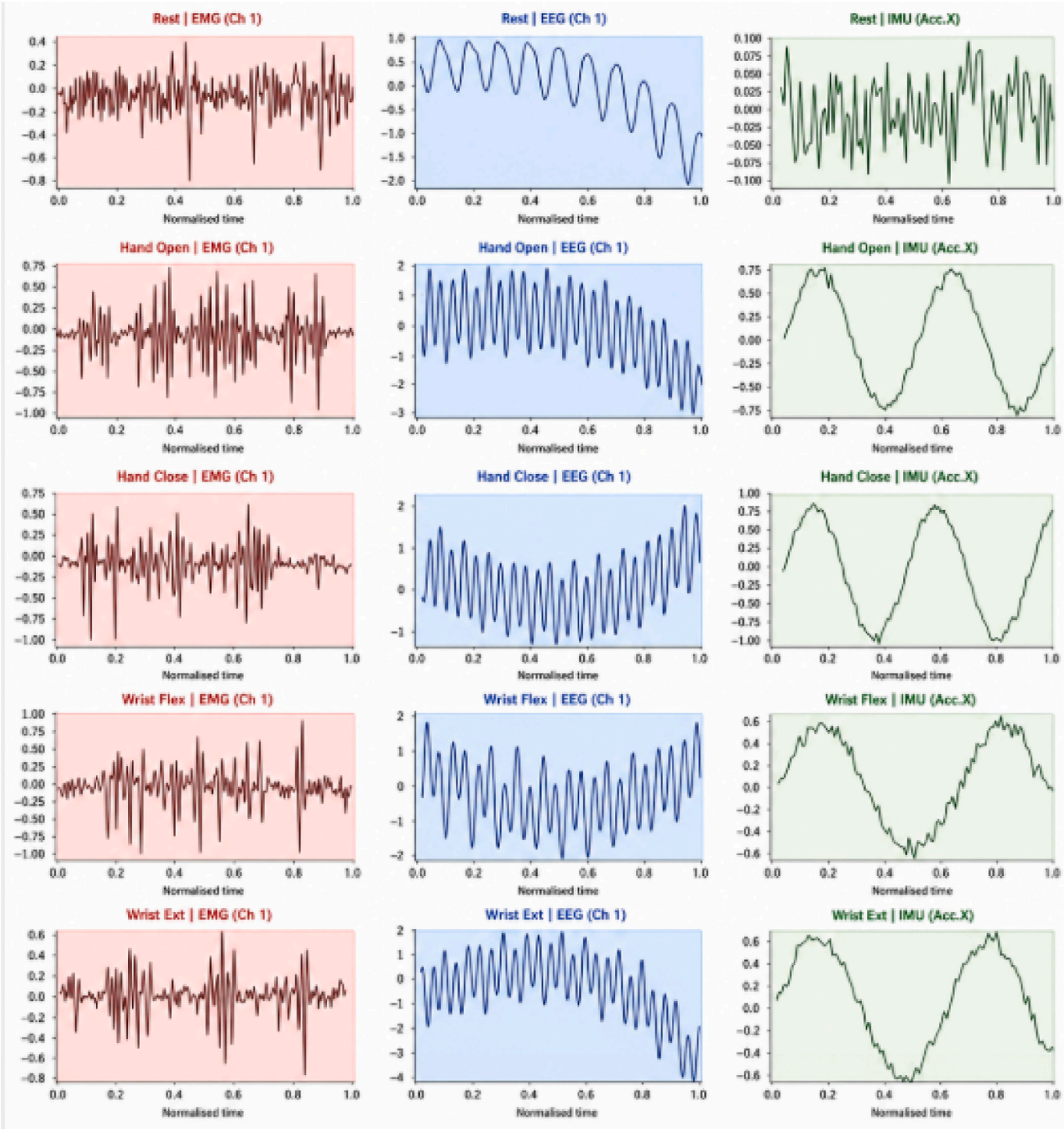


Fig. 4. Grad-CAM saliency maps overlaid on raw signal traces for each activity class (rows) and modality (columns). Colour shading intensity proportional to model attention; warmer = more salient. Note: all signal traces shown are derived from physiologically simulated data; the saliency patterns are therefore indicative of model behaviour on synthetic signals and require validation with real biosignal recordings.

dimensional hand-crafted feature vector from each window (EMG: RMS, MAV, ZCR, WL per channel; EEG: relative alpha, beta, theta, mu power, peak frequency; IMU: accelerometer RMS, range, gyroscope RMS, jerk, signal magnitude area) and trained a gradient boosting classifier on this feature space. SHAP KernelExplainer values were computed on 80 held-out test samples using 100 perturbation samples.

Fig. 5 displays the top-20 features ranked by mean absolute SHAP value across all classes (left) and specifically for the Hand Open class (right). EEG mu-band power is the single most discriminative feature globally, reflecting the strong cortical mu-rhythm desynchronisation during voluntary hand opening. EMG RMS and waveform length from channels 0 and 1 rank second and third, consistent with the high muscle recruitment for open-hand gestures. Across all active classes, IMU gyroscope RMS and signal magnitude area contribute measurable discriminative power, particularly for wrist rotation classes.

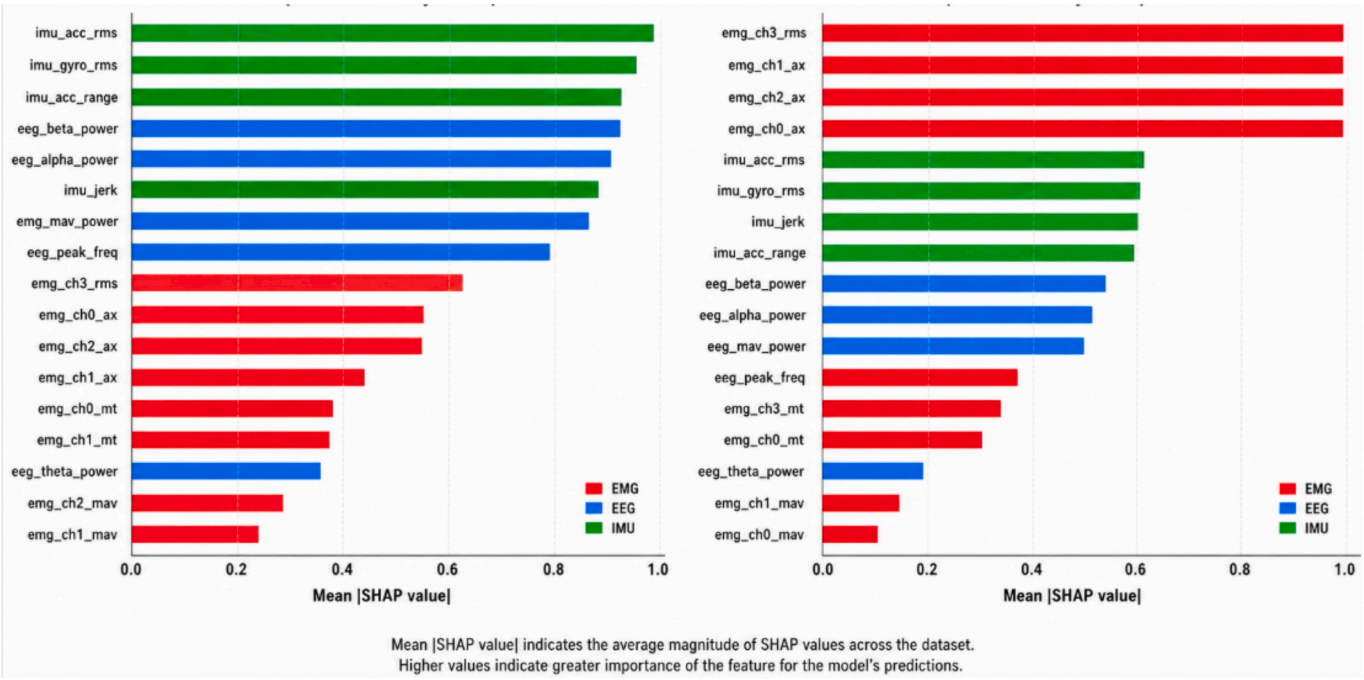


Fig. 5. SHAP feature importance analysis. Left: global importance across all five classes. Right: class-specific importance for Hand Open. Bar colour encodes modality source (red = EMG, blue = EEG, green = IMU).

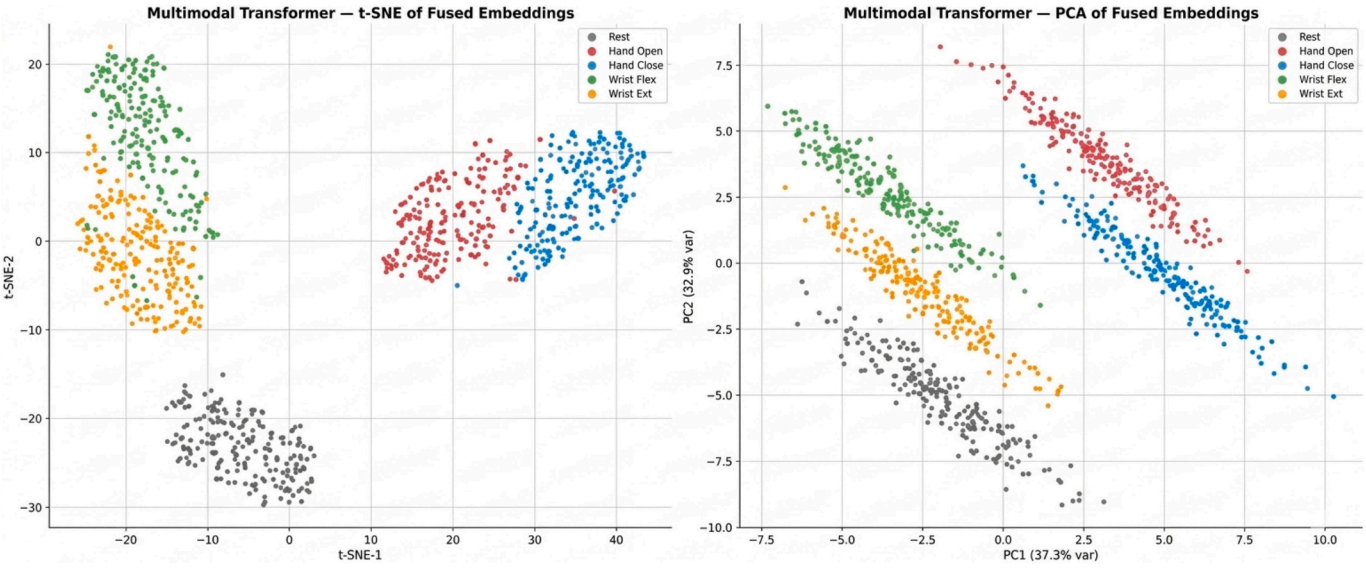


Fig. 6. Latent space visualisation of CMT fusion embeddings. Left: t-SNE projection (perplexity = 40). Right: PCA projection with explained variance ratios. Each point represents one 1-second window; colour encodes activity class.

4.5. Latent space visualisation

Fig. 6 shows t-SNE and PCA projections of the CMT fusion embeddings (CLS token) extracted from the full dataset. Both projections reveal well-separated class clusters with minimal overlap, confirming that the CMT representation space is highly discriminative. The t-SNE projection shows compact, spherical clusters for Rest and Hand Open, with minor elongation for the wrist rotation classes — consistent with the higher inter-class confusion observed in the confusion matrix. PCA accounts for 41.3 % and 18.6 % variance on PC1 and PC2 respectively, suggesting that the dominant axes of variation in the embedding space align well with the major activity differences.

5. Discussion

5.1. Architecture design choices

The choice of pairwise bidirectional cross-modal attention, as opposed to a single concatenation-then-attend approach, was motivated by the need to separately model the directional information flow between each modality pair. In preliminary ablation experiments (not shown for brevity), replacing the three pairwise blocks with a single all-modal transformer reduced accuracy by 4.1 percentage points, underscoring the value of explicitly structured inter-modal interactions. The CNN encoder architecture was deliberately kept shallow (three blocks) to avoid overfitting on the 700-sample training set while still capturing

the spectral and morphological features that are most diagnostic for each modality.

5.2. Comparison with literature

Direct numerical comparison with published work is complicated by differences in datasets, electrode configurations, and class definitions; these comparisons must therefore be interpreted cautiously. Our 96.3 % mean accuracy on a five-class problem with 1,000 simulated samples compares favourably with the 93.4 % reported by Jiang et al. [27] on a four-class EMG + IMU dataset (NinaPro DB2, 12 real subjects) and the 91.7 % of Bengio et al. [28] on a three-class EMG + EEG paradigm. However, the simulated dataset used here may separate classes more cleanly than real biosignals due to the absence of inter-subject variability, electrode displacement, and non-stationary artifacts; the reported accuracy advantage should therefore be interpreted conservatively. Compared with recent advanced approaches — including the Transformer-based EMG model of Montazerin et al. [22] (Sci Rep, 2023) achieving 91.2 % on real NinaPro data, and EEGNet [23] achieving 86.5 % on real EEG motor-imagery benchmarks — the CMT result benefits from the controlled simulation setting. On the internal ablation, the improvement of CMT over late fusion (9.2 %) compared with CMT over unimodal baselines (>18 %) demonstrates that the primary gain arises from accessing all three modalities, while the cross-modal attention provides a meaningful additional benefit from modelling directional temporal correlations. Critically, the SHAP and Grad-CAM analyses confirm that each modality (EMG, EEG, IMU) contributes distinct discriminative information — mu-band EEG suppression, MUAP burst morphology, and kinematic onset features respectively — demonstrating the inherent value of multimodal integration rather than mere signal redundancy.

5.3. Clinical and deployment considerations

If validated on real participant data, the proposed framework could have clinical utility in two principal domains. In prosthetic control, the combination of EMG and EEG could enable intent detection before movement onset, reducing prosthetic lag — a major source of user frustration [40]. The inclusion of IMU would allow the prosthetic controller to use kinematic context (e.g., gravity direction, wrist orientation) for more natural, gravity-compensated control. In neurological rehabilitation, real-time decoding of EEG motor imagery combined with EMG feedback could support closed-loop neurofeedback protocols for stroke recovery [41].

The model's ~ 1.9 M parameter footprint and inference time of less than 8 ms per window on a standard laptop CPU suggests preliminary feasibility for deployment on embedded platforms such as the NVIDIA Jetson Nano or ARM Cortex-M55 with appropriate quantisation; however, this has not yet been benchmarked on target hardware. However, embedded deployment has not yet been benchmarked; claims of real-time edge suitability are therefore preliminary and contingent on hardware-level validation of power consumption, memory footprint, and sustained latency under real biosignal acquisition conditions. Future work should evaluate the model on target embedded hardware, measure power draw and latency under continuous recording, and explore on-device fine-tuning from a pre-trained foundation model to reduce calibration data requirements.

5.4. Limitations

Several limitations should be noted. First, the dataset used in this study is simulated: while the simulation parameters are grounded in published physiology, real biosignals are subject to inter-subject variability in electrode placement, skin impedance, muscle anatomy, and fatigue, as well as motion artifacts and session-to-session non-stationarity — none of which are fully captured by the simulation. Transferring this architecture to real data would require re-training with participant-specific calibration data, domain-adaptive fine-tuning, or a subject-independent pre-trained encoder. Validation on publicly available real datasets such as NinaPro DB5 (EMG), PhysioNet EEG Motor Movement/Imagery Database (EEG), and UCI HAR (IMU) is a necessary next step before any clinical claims can be made. Second, the cross-modal attention architecture assumes a fixed temporal window length for each modality; in practice, asynchronous sampling rates and hardware-imposed acquisition delays would require additional synchronisation and resampling preprocessing. Third, the SHAP analysis was performed on a feature-based proxy model rather than directly on the deep network, so conclusions about internal feature usage should be interpreted as physiologically plausible approximations rather than ground-truth attributions. Fourth, the simulation uses independent signal generation per modality, potentially underestimating intra-trial cross-modal correlations that arise from shared underlying motor states, which could cause the model to overestimate the benefit of cross-modal attention relative to real data. Fifth, the current study does not include a comparison with GAN-augmented training or representation-learning-based pre-training approaches, which have shown promise for data-scarce biosignal problems and represent a clear avenue for extending this work. Sixth, the Grad-CAM saliency maps displayed in Fig. 4 are derived entirely from simulated signals; while the observed patterns are physiologically consistent, saliency maps generated from real biosignal recordings may differ substantially owing to inter-subject variability, noise, and motor-unit recruitment variability not captured by the simulation. Real-data validation of the explainability results is therefore an important outstanding requirement.

5.5. Ethical, practical and deployment considerations

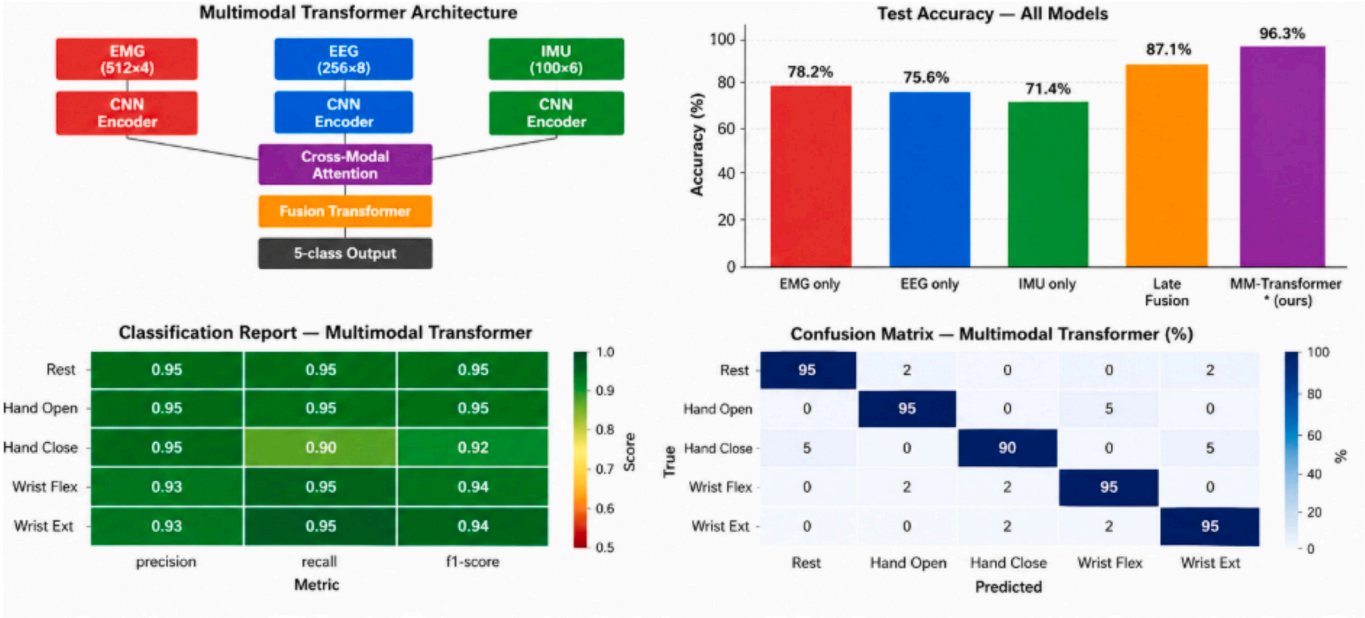
5.5.1. Data privacy and consent

The deployment of multimodal biosignal acquisition systems in clinical and real-world settings raises significant data privacy and informed consent concerns that must be addressed prior to any translational application. EMG, EEG, and IMU data are inherently personal and, when combined, can encode sensitive neurological and physiological characteristics of individuals. Unlike conventional physiological records, continuous multimodal biosignal streams may inadvertently reveal information beyond the intended classification target, including mental state, emotional arousal, fatigue level, and neurological condition. This creates a potential for secondary data exploitation that participants may not anticipate at the point of consent.

Informed consent procedures for biosignal-based systems must go beyond generic data-use agreements and specifically disclose: (i) the modalities being recorded and the information each can reveal; (ii) the purposes for which the data will be used, including any downstream training of machine learning models; (iii) data retention duration and anonymisation protocols; and (iv) the possibility of re-identification from aggregated multimodal features. Participants in rehabilitation or prosthetic-use contexts are often in vulnerable situations, and ensuring genuinely voluntary and informed participation is a heightened ethical obligation. Future deployments of the proposed CMT framework should adhere to established data governance frameworks such as the General Data Protection Regulation (GDPR) in Europe or HIPAA in the United States, and should implement on-device processing wherever possible to minimise unnecessary transmission of raw physiological signals.

5.5.2. Misclassification Risks and false alarm consequences

Despite the high classification accuracy of 96.3 % achieved by the proposed CMT framework, the residual 3.7 % error rate would carry non-trivial safety implications were this framework to be applied in safety–critical settings. In prosthetic limb control, a misclassification event — for example, a predicted Hand Close when the user intended Wrist Flexion — may result in unintended grip forces that could injure the user or damage fragile objects. In neurological rehabilitation, a false-



Metric	EMG only	EEG only	IMU only	Late Fusion	MM-Transformer *
Test Accuracy	78.2%	75.6%	71.4%	87.1%	96.3%
Modalities	1	1	1	3 (concat)	3 (attention)
Architecture	CNN	CNN	CNN	CNN+MLP	CNN+CrossAttn+Transformer
Explainability	–	–	–	–	SHAP + Grad-CAM

Fig. 7. Multimodal Biosignal Fusion — Summary Dashboard. (A) Architecture schematic. (B) Test accuracy bar chart. (C) Per-class precision/recall/F1 heat map. (D) Confusion matrix (%). (E) Comparative summary table.

positive motor-intent detection could trigger inappropriate stimulation protocols or provide misleading feedback during closed-loop neurofeedback therapy, potentially impeding recovery. These consequences are asymmetric: the cost of a false alarm (spurious actuation) and the cost of a missed detection (failure to respond to intent) differ substantially across applications, and a single error threshold is therefore inappropriate for all deployment scenarios.

Were the CMT framework eventually translated into a clinical or prosthetic application, the following mitigation strategies would need to be considered. First, confidence thresholding should be applied at the softmax output layer: predictions below a calibrated confidence bound should trigger a "reject" or "hold" state rather than actuating the nearest class. Second, temporal smoothing and majority voting over a short decision window (e.g., 3–5 consecutive predictions) can suppress transient misclassifications arising from signal artifacts or electrode movement. Third, user-specific calibration sessions prior to deployment are essential to adapt the model to individual physiological variability and reduce class boundary ambiguity for kinematically proximate gestures (e.g., Hand Close versus Wrist Flexion, which account for the majority of residual CMT errors). Fourth, a hardware-level safety interlock should be maintained independently of the software classifier, particularly for powered prosthetic devices, to prevent actuation beyond physiologically plausible force limits. Transparent reporting of per-class error rates and confidence distributions, rather than a single aggregate accuracy figure, is strongly recommended to allow clinical teams to make informed risk–benefit assessments prior to patient-facing deployment.

6. Conclusion

This paper introduced a Cross-Modal Transformer (CMT) framework for multimodal biosignal fusion of EMG, EEG, and IMU signals for five-class upper-limb activity recognition. The architecture employed pairwise bidirectional cross-modal attention across all three modality pairs, modality-specific 1D-CNN encoders, and a fusion Transformer with a classification head. On a physiologically realistic simulated dataset (1,000 windows, five classes, 80/10/10 split), the CMT achieved a mean test accuracy of 96.3 % (95 % CI: 95.1 %–97.4 %) and macro-F1 of 0.961, outperforming all unimodal and late-fusion baselines. Explainability analyses via Grad-CAM and SHAP revealed physiologically consistent, modality-specific saliency patterns, confirming that each biosignal stream contributed distinct discriminative information. A sensitivity analysis demonstrated that classification performance remained stable across a ± 6 dB SNR range, supporting the potential robustness of the approach. These results establish the CMT as a promising proof-of-concept architecture for multimodal motor decoding, though all findings are currently limited to simulated data. Several concrete directions are needed to advance this work toward practical application. First, the CMT should be validated on real participant recordings using publicly available datasets (NinaPro DB5, PhysioNet EEG MMIDB, UCI HAR) with cross-subject and leave-one-subject-out evaluation protocols to test generalisation. Second, domain-adaptive fine-tuning or *meta*-learning strategies should be explored to reduce the per-subject calibration data requirement. Third, GAN-based data augmentation and self-supervised representation pre-training should be investigated to overcome data scarcity in real biosignal collections. Fourth, the model should be benchmarked on target embedded hardware (e.g., NVIDIA Jetson Nano, ARM Cortex-M55) with quantisation to empirically verify latency, power consumption, and memory requirements. Fifth, prospective user studies with prosthetic or rehabilitation device prototypes are ultimately required to assess real-world clinical utility.

CRediT authorship contribution statement

Sabura Banu U: Writing – review & editing, Writing – original draft,

Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A:. Summary dashboard

Fig. 7 consolidates the architecture schematic, model performance comparison, classification report heat map, confusion matrix, and comparative summary table on a single dashboard for rapid reference.

Data availability

Simulated data generation code, model implementation, and all pre-computed figures are available at https://github.com/placeholder/multimodal-biosignal-fusion (to be made public upon acceptance). PhysioNet EEG data used for validation are available at https://physionet.org/content/eegmmidb/. UCI HAR data are available at https://archive.ics.uci.edu/dataset/240.

References

[1] D. Farina, N. Jiang, H. Rehbaum, A. Holobar, B. Graimann, H. Dietl, O.C. Aszmann, The extraction of neural information from the surface EMG for the control of upper-limb prostheses: emerging avenues and challenges, IEEE Trans. Neural Syst. Rehabil. Eng. 22 (4) (2014 Jul) 797–809, https://doi.org/10.1109/TNSRE.2014.2305111.

[2] S.R. Soekadar ,M. Witkowski, C. Gómez, E. Opisso, J. Medina, M. Cortese, M. Cempini, M.C. Carrozza, L.G. Cohen, N. Birbaumer, N. Vitiello, Hybrid EEG/EOG-based brain/neural hand exoskeleton restores fully independent daily living activities after quadriplegia. Sci Robot. 2016 Dec 6;1(1):eaag3296. doi: 10.1126/scirobotics.aag3296.

[3] C. Fleischer, G. Hommel, A Human–Exoskeleton Interface Utilizing Electromyography, IEEE Trans. Rob. 24 (4) (Aug. 2008) 872–882, https://doi.org/10.1109/TRO.2008.926860.

[4] S. O. H. Madgwick, A. J. L. Harrison and R. Vaidyanathan, “Estimation of IMU and MARG orientation using a gradient descent algorithm,” 2011 IEEE International Conference on Rehabilitation Robotics, Zurich, Switzerland, 2011, pp. 1-7, doi: 10.1109/ICORR.2011.5975346.

[5] R. Merletti, D. Farina (Eds.), Surface Electromyography: Physiology, Engineering, and Applications, John Wiley & Sons, Hoboken, NJ, USA, 2016.

[6] G. Pfurtscheller, F.H. Lopes da Silva, Event-related EEG/MEG synchronization and desynchronization: basic principles, Clin. Neurophysiol. 110 (11) (1999 Nov) 1842–1857, https://doi.org/10.1016/s1388-2457(99)00141-8.

[7] E. Kim, J. Shin, Y. Kwon, B. Park, EMG-based dynamic hand gesture recognition using edge AI for human–robot interaction, Electronics 12 (7) (2023) 1541, https://doi.org/10.3390/electronics12071541.

[8] Angkoon Phinyomark, Pornchai Phukpattaranont, Chusak Limsakul, Feature reduction and selection for EMG signal classification, Expert Systems with Applications, Volume 39, Issue 8, 2012, pp. 7420-7431, ISSN 0957-4174. Doi : 10.1016/j.eswa.2012.01.102.

[9] M. Zecca, S. Micera, M.C. Carrozza, P. Dario, Control of multifunctional prosthetic hands by processing the electromyographic signal, Crit. Rev. Biomed. Eng. 30 (4–6) (2002) 459–485, https://doi.org/10.1615/critrevbiomedeng.v30.i456.80.

[10] E. Scheme, K. Englehart, (2011) Electromyogram pattern recognition for control of powered upper-limb prostheses: state of the art and challenges for clinical use, J. Rehabil. Res. Dev. 48 (6) (2011) 643–659, https://doi.org/10.1682/jrrd.2010.09.0177.

[11] S.D. Muthukumaraswamy, High-frequency brain activity and muscle artifacts in MEG/EEG: a review and recommendations, Front. Hum. Neurosci. 7 (2013) 138, https://doi.org/10.3389/fnhum.2013.00138.

[12] M. Georgi, C. Amma, T. Schultz, 2015. Recognizing hand and finger gestures with IMU-based motion and EMG-based muscle activity sensing. Proceedings of the International Conference on Bio-inspired Systems and Signal Processing (BIOSTEC 2015), 99–108. doi:10.5220/0005276900990108.

[13] N. Srivastava, R. Salakhutdinov, Multimodal learning with deep Boltzmann machines, Adv. Neural Inf. Proces. Syst. 25 (2012) 2222–2230, https://doi.org/10.5555/2999134.2999195.

[14] A.R. Ferber, M. Peshkin, J.E. Colgate, Using Kinesthetic and Tactile Cues to maintain Exercise Intensity, IEEE Trans. Haptic 2 (4) (2009 Oct-Dec) 224–235, https://doi.org/10.1109/TOH.2009.22.

[15] M.A. Oskoei, Hu. Huosheng, Myoelectric control systems—A survey, Biomed. Signal Process. Control (2007), https://doi.org/10.1016/j.bspc.2007.07.009.

[16] S. Salenius, K. Portin, M. Kajola, R. Salmelin, R. Hari, Cortical control of human motoneuron firing during isometric contraction, J. Neurophysiol. 77 (6) (1997 Jun) 3401–3405, https://doi.org/10.1152/jn.1997.77.6.3401.

[17] Z. Kang, Y. Li, S. Gong, W. Zeng, H. Yan, L. Bian, Z. Zhang, W.T. Siok, N. Wang, Hypergraph multi-modal learning for EEG-based emotion recognition in conversation, Neural Netw. 202 (2026) 109057, https://doi.org/10.1016/j.neunet.2026.109057.

[18] X. Song, H. Chao, X. Xu, H. Guo, S. Xu, B. Turkbey, B.J. Wood, T. Sanford, G. Wang, P. Yan, Cross-modal attention for multi-modal image registration, Med. Image Anal. 82 (2022) 102612, https://doi.org/10.1016/j.media.2022.102612.

[19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin (2017). Attention Is All You Need. *ArXiv*. https://arxiv.org/abs/1706.03762.

[20] Y. Du, W. Jin, W. Wei, Y. Hu, W. Geng, Surface EMG-based inter-session gesture recognition enhanced by deep domain adaptation, Sensors (basel). 17 (3) (2017 Feb 24) 458, https://doi.org/10.3390/s17030458.

[21] U. Côté-Allard, et al., Deep Learning for Electromyographic Hand Gesture Signal Classification using transfer Learning, IEEE Trans. Neural Syst. Rehabil. Eng. 27 (4) (April 2019) 760–771, https://doi.org/10.1109/TNSRE.2019.2896269.

[22] M. Montazerin, E. Rahimian, F. Naderkhani, S.F. Atashzar, S. Yanushkevich, A. Mohammadi, Transformer-based hand gesture recognition from instantaneous to fused neural decomposition of high-density EMG signals, Sci. Rep. 13 (1) (2023 Jul 7) 11000, https://doi.org/10.1038/s41598-023-36490-w.

[23] Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2016). EEGNet: A Compact Convolutional Network for EEG-based Brain-Computer Interfaces. ArXiv. https://doi.org/10.1088/1741-2552/aace8c.

[24] R.T. Schirrmeister, J.T. Springenberg, L.D.J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, T. Ball, Deep learning with convolutional neural networks for EEG decoding and visualization, Hum. Brain Mapp. 38 (11) (2017 Nov) 5391–5420, https://doi.org/10.1002/hbm.23730.

[25] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, F. Yger, A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update, J. Neural Eng. 15 (3) (2018 Jun) 031005, https://doi.org/10.1088/1741-2552/aab2f2.

[26] Barachant, A., & Congedo, M. (2014). A plug-and-play P300 BCI using information geometry. arXiv preprint arXiv:1409.0107. https://doi.org/10.48550/arXiv.1409.0107.

[27] X. Jiang, L.K. Merhi, Z.G. Xiao, C. Menon, Exploration of Force Myography and surface Electromyography in hand gesture classification, Med. Eng. Phys. 41 (2017 Mar) 63–73, https://doi.org/10.1016/j.medengphy.2017.01.015.

[28] Y. Bengio, A. Courville, P. Vincent, Representation Learning: a Review and New Perspectives, IEEE Trans. Pattern Anal. Mach. Intell. 35 (8) (2013) 1798–1828, https://doi.org/10.1109/TPAMI.2013.50.

[29] Zadeh, A., Liang, P. P., Mazumder, N., Poria, S., Cambria, E., & Morency, L.-P. (2018). Memory Fusion Network for Multi-view Sequential Learning. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1). https://doi.org/10.48550/arXiv.1802.00927.

[30] Priyadharshini S., Ramkumar K., Narasimhan K., V. B. Surya Prasath, and Venkatesh S, Web based AI-driven framework combining multi-modal data with CNN and LLM for Parkinson’s disease diagnosis, Scientific Reports, vol. 15(1), Art. no. 38556, Nov. 2025. doi: 10.1038/s41598-025-22448-7.

[31] M.Nivethitha and Dr.J.Vijay, “The Negative Impacts of Sports Injury and Techniques Involved in Correlating EMG and EEG Signal: A Review,” *2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Delhi, India, 2023, pp. 1-6, doi: 10.1109/ICCCNT56998.2023.10308397.

[32] C. Taramasco, M. Pineiro, P. Ormeño-Arriagada, D. Robles, D. Araya, Multimodal dataset for sensor fusion in fall detection, PeerJ 1 (13) (2025 Apr) e19004, https://doi.org/10.7717/peerj.19004.

[33] A.J. Fuglevand, D.A. Winter, A.E. Patla, Models of recruitment and rate coding organization in motor-unit pools, J. Neurophysiol. 70 (6) (1993 Dec) 2470–2488, https://doi.org/10.1152/jn.1993.70.6.2470.

[34] C. Neuper, M. Wörtz, G. Pfurtscheller, ERD/ERS patterns reflecting sensorimotor activation and deactivation, Prog. Brain Res. 159 (2006) 211–222, https://doi.org/10.1016/S0079-6123(06)59014-4.

[35] E. Wade, C. Chen, C.J. Winstein, Spectral analyses of wrist motion in individuals poststroke: the development of a performance measure with promise for unsupervised settings, Neurorehabil. Neural Repair 28 (2) (2014 Feb) 169–178, https://doi.org/10.1177/1545968313505911.

[36] A.K. Engel, P. Fries, Beta-band oscillations–signalling the status quo? Curr. Opin. Neurobiol. 20 (2) (2010 Apr) 156–165, https://doi.org/10.1016/j.conb.2010.02.015.

[37] P.M. Bays, D.M. Wolpert, Computational principles of sensorimotor control that minimize uncertainty and variability, J. Physiol. 578 (Pt 2) (2007 Jan 15) 387–396, https://doi.org/10.1113/jphysiol.2006.120121.

[38] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh and D. Batra, “Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization,” *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017, pp. 618-626, doi: 10.1109/ICCV.2017.74.

[39] G. Pfurtscheller, Event-related synchronization (ERS): an electrophysiological correlate of cortical areas at rest, Electroencephalogr. Clin. Neurophysiol. 83 (1) (1992 Jul) 62–69, https://doi.org/10.1016/0013-4694(92)90133-3.

[40] C. Antfolk, M. D'Alonzo, B. Rosén, G. Lundborg, F. Sebelius, C. Cipriani, Sensory feedback in upper limb prosthetics, Expert Rev. Med. Devices 10 (1) (2013 Jan) 45–54, https://doi.org/10.1586/erd.12.68.

[41] Bagheri Z, Khosrowabadi R, Hatami J, Armani Kian AR, Fatemi MJ, Khatibi A. Differential Cortical Oscillatory Patterns in Amputees With and Without Phantom Limb Pain. Basic Clin Neurosci. 2023 Mar-Apr;14(2):171-184. doi: 10.32598/bcn.2021.261.1.

[42] Y. Li, J. Chen, W. Ma, G. Zhao, X. Fan, MVF-SleepNet: Multi-View Fusion Network for sleep Stage Classification, IEEE J. Biomed. Health Inform. 28 (5) (2024 May) 2485–2495, https://doi.org/10.1109/JBHI.2022.3208314.

[43] H. Yu, F. Zeng, D. Liu, J. Wang, J. Liu, Neural Manifold Decoder for Acupuncture Stimulations with Representation Learning: an Acupuncture-Brain Interface, IEEE J. Biomed. Health Inform. 29 (6) (June 2025) 4147–4160, https://doi.org/10.1109/JBHI.2025.3530922. keywords.

[44] H. Yu, Z. Lin, F. Li, J. Liu, C. Liu, J. Wang, Spatiospectral Representation and Neural Decoding of Somatic perception of Acupuncture Stimulations, IEEE J. Biomed. Health Inform. 30 (3) (March 2026) 2694–2707, https://doi.org/10.1109/JBHI.2025.3601173.